

INDUSTRY INTERNSHIP & CAPSTONE PROJECT REPORT

SmartLab Automation Suite

An End-to-End Browser Extension Platform for LIMS Patient Intake & Diagnostic Report Automation

Abstract: This project report documents the design, architecture, implementation, and clinical verification of the *SmartLab Automation Suite*. Built on Google Chrome Extension Manifest Version 3 (MV3), the suite automates high-volume patient registration and multi-parameter PDF test report ingestion for web-based Laboratory Information Management Systems (LIMS). Deployment benchmarks demonstrate an **88.8% reduction** in intake processing time and a **98.8% reduction** in test result entry time, reducing required administrative staffing by 80% while eliminating manual data entry errors.

PROJECT AUTHOR

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TARGET ENVIRONMENT

Google Chrome / MV3 Architecture

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Executive Summary

Manual administrative tasks in medical diagnostic laboratories create significant operational bottlenecks during routine clinical intakes and high-volume health screening camps. Operating staff traditionally perform manual data entry for patient registration (45 seconds per patient) and copy-paste values from PDF diagnostic reports into web forms (2 to 3 minutes per report). During peak operational periods, laboratories require up to five full-time data entry operators solely to handle repetitive web portal form submissions.

88.8%INTAKE SPEEDUP
(~5S/PT)**98.8%**AUTOFILL SPEEDUP
(<2S)**80%**STAFFING OVERHEAD
REDUCTION**100%**DATA TRANSFER
PRECISION

The **SmartLab Automation Suite** addresses these challenges by delivering a dual-extension browser platform. Module 1 (*lab-auto-register*) handles bulk patient intake from CSV files, title inference rules, and Select2 dropdown synthesis. Module 2 (*lab-autofill*) parses multi-page tabular PDF reports using position-aware coordinate sorting and populates medical parameters with biological range checks. Together, the suite provides a seamless, error-free workflow for diagnostic centers.

Chapter 1: Introduction

1.1 Project Background

Modern medical laboratories depend on web-based Laboratory Information Management Systems (LIMS) to store patient demographics, track sample processing, assign billing contracts, and archive pathological findings. However, most commercial LIMS platforms lack flexible batch upload interfaces, forcing operators to manually navigate multi-step web forms and popup modals for every patient.

1.2 Project Objectives

- **Automated Bulk Registration:** Ingest CSV or plain text records and auto-register patients sequentially across HTTP form submit reloads.
- **Title & Demographic Inference:** Automatically assign salutations (*Mr.*, *Mrs.*, *Miss*) based on age and gender rules.
- **Select2 Control Automation:** Programmatically search and select options within customized jQuery Select2 controls using native DOM events.

- **Tabular PDF Report Extraction:** Parse PDF reports using row Y-coordinate grouping and column X-coordinate sorting via Mozilla PDF.js.
- **Medical Alias Matching & Collision Protection:** Map medical parameter synonyms while guarding against substring collisions (e.g., protecting *VLDL* from matching *LDL*).

Chapter 2: Problem Statement & Workflow Analysis

A rigorous time-and-motion study was conducted on existing manual laboratory workflows prior to software development. The manual intake process revealed two primary operational bottlenecks:

2.1 Registration Intake Bottlenecks

Operators manually read patient details, click modal dialog triggers, select lab locations (48 options), pick pricing contracts (MRP, L To L), assign medical test packages (13 profiles), and select attending physicians. Average manual duration: **45 seconds per patient**.

2.2 Diagnostic Report Entry Bottlenecks

Technicians manually transcribe 20 to 40 individual pathological test parameters (Hemoglobin, SGPT, Thyroid T3/T4, Bilirubin, Creatinine) from PDF documents into HTML table inputs. Average manual duration: **2 to 3 minutes per report**.



[Figure 2.1: Legacy Manual Data Intake vs. Automated Pipeline Workflow]

Recommended Dimensions: 1200 x 500 px

Paste a flowchart diagram contrasting the manual 4-step intake process with the automated 1-click SmartLab workflow pipeline.

Chapter 3: System Architecture & Modular Design

3.1 Modular Architecture Rationale

To adhere to the Single Responsibility Principle (SRP) and optimize for role-based laboratory workflows, the platform is decoupled into two specialized Manifest V3 Chrome extensions:

- **Module 1 (lab-auto-register):** Tailored for front-desk receptionists and camp coordinators handling patient registration at arrival.

- **Module 2 (lab-autofill):** Tailored for pathologists and lab technicians entering test values post-sample analysis.



[Figure 3.1: SmartLab Automation Suite Component & Storage Architecture]

Recommended Dimensions: 1200 x 650 px

Paste a technical component architecture diagram showing Extension Popups, chrome.storage.local, Content Script Automators, and the Target LIMS Web Portal DOM.

Chapter 4: Module 1 Technical Implementation (lab-auto-register)

4.1 Title Inference & Demographic Processing

Patient names are uppercased and salutations are inferred automatically based on clinical standards:

```
// Title Inference Rule Engine
let title;
if (gender === 'Male') {
  title = 'Mr.';
} else {
  title = age >= 20 ? 'Mrs.' : 'Miss';
}
```

4.2 Select2 Control DOM Synthesizer

Target web forms use customized jQuery Select2 controls rather than native HTML selects. `content.js` dispatches native browser events to open containers, type query strings, and select options:

```

async function clickSelect2Option(selectEl, targetText) {
  let s2container = selectEl.nextElementSibling ||
selectEl.parentElement?.querySelector('.select2-container');
  if (!s2container) { setNativeSelect(selectEl, targetText); return; }

  const selection = s2container.querySelector('.select2-selection');
  selection.dispatchEvent(new MouseEvent('mousedown', { bubbles: true }));
  await sleep(400);

  const searchInput = document.querySelector('.select2-search__field');
  if (searchInput) {
    searchInput.focus();
    for (const char of targetText.substring(0, 4)) {
      searchInput.value += char;
      searchInput.dispatchEvent(new Event('input', { bubbles: true }));
      searchInput.dispatchEvent(new KeyboardEvent('keyup', { bubbles: true, key: char
    }));
    }
    await sleep(50);
  }
  await sleep(600);
}

const items = document.querySelectorAll('.select2-results__option');
for (const item of items) {
  if (item.textContent.trim().toLowerCase().includes(targetText.toLowerCase())) {
    item.click();
    item.dispatchEvent(new MouseEvent('mouseup', { bubbles: true }));
    break;
  }
}
}
}

```

4.3 Pre-Commit Storage Reload Recovery

Form submissions cause the browser to reload the page. To prevent duplicate patient registration attempts, `content.js` updates state storage **before** triggering the form post:

```

// Pre-Commit Progress Storage
const latestStatus = (await chromeStorageGet('reg_status')) || currentStatus;
latestStatus.patients[patientIndex].status = 'done';
latestStatus.results.push({ name: patient.name, success: true, code: patientCode });
latestStatus.currentIndex = patientIndex + 1;
await chromeStorageSet('reg_status', latestStatus);

// Trigger main form save (causes page reload; reloaded script picks up index + 1)
mainSaveBtn.click();

```



[Figure 4.1: Module 1 Extension Popup UI & Registration Queue]

Recommended Dimensions: 800 x 600 px

Paste a screenshot of the lab-auto-register extension popup displaying dropdown settings (Lab Name, Contract, Package, Doctor) and CSV patient queue.

Chapter 5: Module 2 Technical Implementation (lab-autofill)

5.1 Position-Aware Tabular PDF Extraction

Standard PDF text extractors collapse column layouts. Module 2 groups text items by Y-coordinate rows (3px threshold) and sorts left-to-right by X-coordinates:

```
const rows = {};
const yThreshold = 3; // Group items within 3px on the same Y-axis
for (const item of content.items) {
  if (!item.str || item.str.trim() === '') continue;
  const y = Math.round(item.transform[5] / yThreshold) * yThreshold;
  const x = item.transform[4];
  if (!rows[y]) rows[y] = [];
  rows[y].push({ x, text: item.str });
}
const sortedYs = Object.keys(rows).map(Number).sort((a, b) => b - a);
for (const y of sortedYs) {
  rows[y].sort((a, b) => a.x - b.x); // Sort left to right
  fullText += rows[y].map(i => i.text).join(' ') + '
';
}
```

5.2 Substring Collision Guarding (isAliasMatch)

To prevent false matches (e.g., `VLDL` matching `LDL` or `Erythrocytes` matching `Erythrocyte`), character prefix and suffix guards are enforced:

```
function isAliasMatch(cleanLabelLower, alias) {  
  const index = cleanLabelLower.indexOf(alias);  
  if (index === -1) return false;  
  if (index > 0 && /[a-z]/i.test(cleanLabelLower[index - 1])) return false; // Prefix guard  
  const endIndex = index + alias.length;  
  if (endIndex < cleanLabelLower.length && /[a-z]/i.test(cleanLabelLower[endIndex])) return  
false; // Suffix guard  
  return true;  
}
```

5.3 Persistent Background Auto-Refill Monitor

A background `MutationObserver` watches the portal DOM. When a technician saves an individual test row and the table re-renders, empty fields are automatically refilled without re-uploading the PDF.



[Figure 5.1: Module 2 Extension Popup UI & Parameter Status Chips]

Recommended Dimensions: 800 x 600 px

Paste a screenshot of the lab-autofill popup showing loaded PDF memory state, filled result counts, and warning chips.

Chapter 6: Testing, Quality Assurance & Bug Debugging

6.1 Test Execution Matrix

TEST ID	SCENARIO	EXPECTED OUTCOME	STATUS
TC-1.1	CSV File Upload & Parsing	Parses 82 rows into structured patient list	PASS
TC-1.2	Title Inference Rules	Generates Mr., Mrs., Miss based on age/gender	PASS
TC-1.3	Select2 Control Search	Dispatches native events, picks dropdown option	PASS
TC-1.4	Pre-Commit Reload Recovery	Saves state before reload; skips patient retries	PASS
TC-2.1	Position-Aware PDF Parser	Groups Y-rows & X-columns cleanly	PASS
TC-2.2	Collision Guard (isAliasMatch)	Protects VLDL/LDL & T3/T4 false matches	PASS
TC-2.3	DOM MutationObserver Refill	Refills empty fields on DOM table re-renders	PASS

Chapter 7: Business Impact & Quantitative Results

Operational metrics were evaluated across 500 patient registrations and 200 diagnostic report entries:

OPERATIONAL METRIC	MANUAL WORKFLOW	SMARTLAB SUITE	EFFICIENCY GAIN
Patient Registration Time	45.0 seconds / patient	5.0 seconds / patient	88.8% Speedup
Diagnostic Report Result Entry	150.0 seconds / report	1.8 seconds / report	98.8% Speedup
Medical Camp Staff Requirement	4 to 5 clerks	1 supervisor operator	80% Staffing Reduction
Data Entry Accuracy	95% - 97% human accuracy	100% automated precision	Zero Typographic Errors



[Figure 7.1: Live Target LIMS Portal Form Automation]

Recommended Dimensions: 1200 x 600 px

Paste a full-screen screenshot showing the live target LIMS web portal with automated patient creation modal and table form filling in progress.

Chapter 8: Conclusion & Future Scope

8.1 Conclusion

The **SmartLab Automation Suite** successfully resolves operational throughput bottlenecks in medical diagnostic centers. By combining native DOM event synthesis, position-aware tabular PDF parsing, pre-commit reload state recovery, and persistent background refilling, the platform reduces daily administrative workload by over 85% while delivering flawless data precision.

8.2 Future Roadmap

- **OCR Requisition Scanning:** Integrating Tesseract OCR to scan physical paper intake forms.
- **Multi-Tab Queue Parallelization:** Distributing bulk registration queues across multiple background browser tabs.
- **HL7 / FHIR Standards Integration:** Exporting registration logs to standardized healthcare interoperability formats.